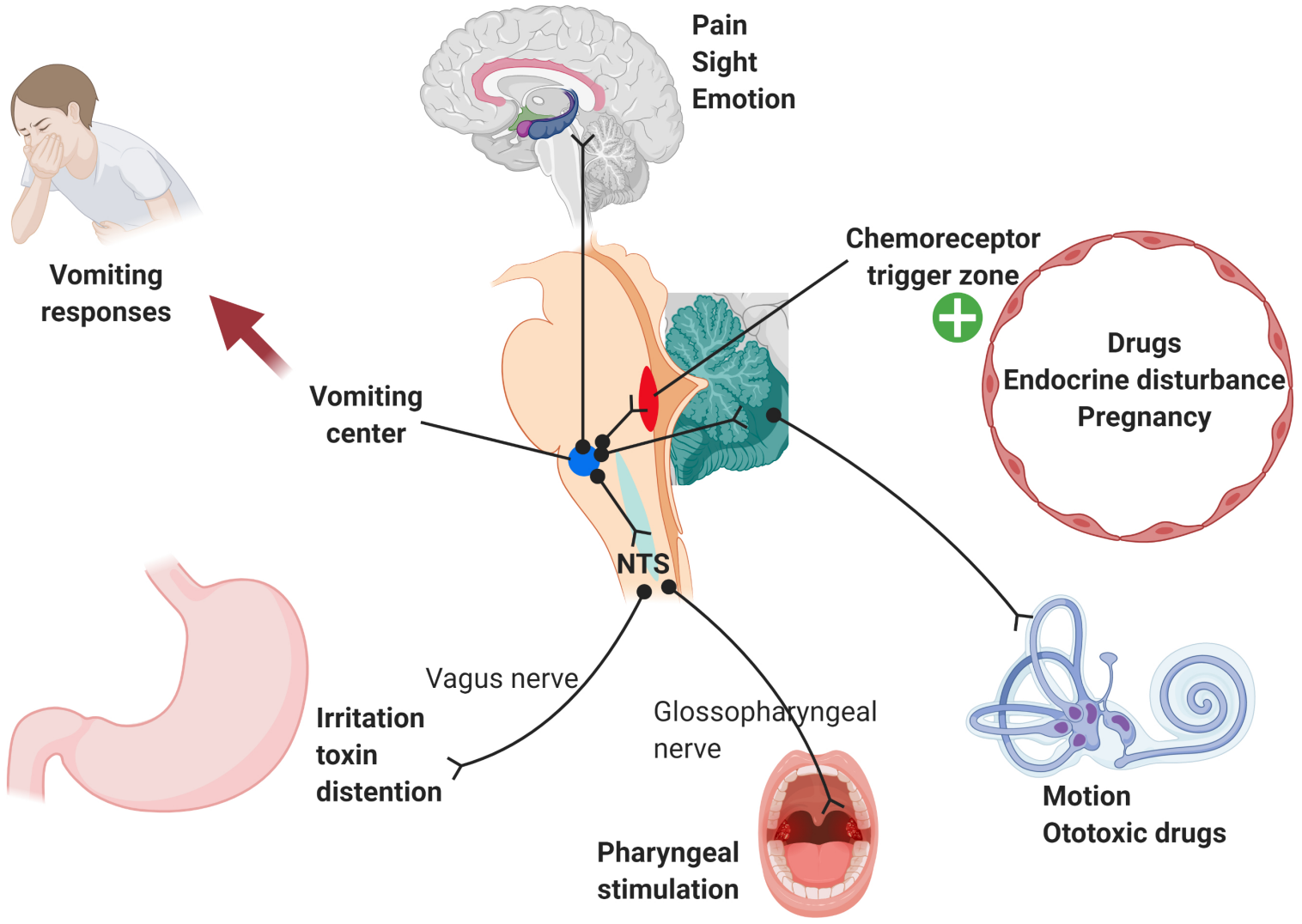
Nausea and vomiting

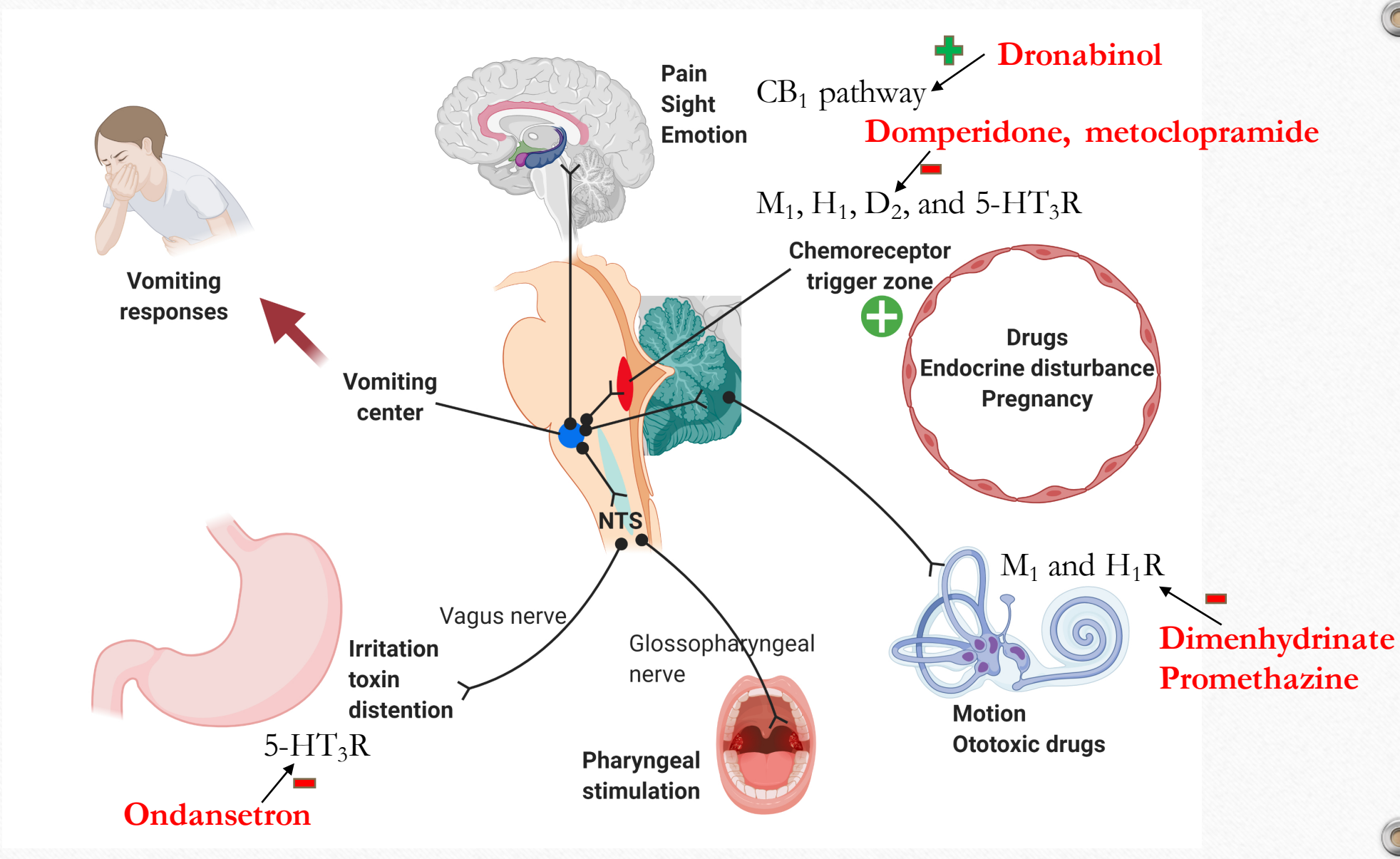
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Vomiting mechanism

- Retrograde contractions → expulsion of gut contents
- Contraction of inspiratory thoracic and abdominal wall muscles → high intra-thoracic and intra-abdominal pressure → evacuate stomach
- Gastric cardia herniates above the diaphragm, and the larynx moves upward → propel the vomitus


Pain
Sight
Emotion
Vomiting responses
Chemoreceptor trigger zone
Drugs
Endocrine disturbance
Pregnancy
Vomiting center
NTS
Vagus nerve
Irritation
toxin
distention
Glossopharyngeal nerve
Pharyngeal stimulation
Motion
Ototoxic drugs


Dronabinol
CB1 pathway
Pain
Sight
Emotion
Domperidone, metoclopramide
M1, H1, D2, and 5-HT3R
Chemoreceptor
trigger zone
Drugs
Endocrine disturbance
Pregnancy
Vomiting
responses
Vomiting
center
NTS
M1 and H1R
Dimenhydrinate
Promethazine
Vagus nerve
Irritation
toxin
distention
Glossopharyngeal
nerve
Motion
Ototoxic drugs
5-HT3R
Pharyngeal
stimulation
Ondansetron

Causes of nausea/vomiting

- **GI related**
- Infection: gastroenteritis, food poisoning
- Obstruction
- Inflammation: cholecystitis, pancreatitis, gastritis, hepatitis
- Altered sensorimotor function: gastroparesis, GERD, cyclic vomiting syndrome

Causes of nausea/vomiting

- **Non-GI related**
- Myocardial infarction
- CNS: cerebral hemorrhage, brain tumor
- Psychiatric: anorexia nervosa, depression
- Inner ear: motion sickness, labyrinthitis

Causes of nausea/vomiting

- **Medications**: chemotherapy, antibiotics, oral contraception
- **Metabolic/endocrine problems**
- Pregnancy
- Uremia
- Ketoacidosis
- Adrenal insufficiency

Thank you